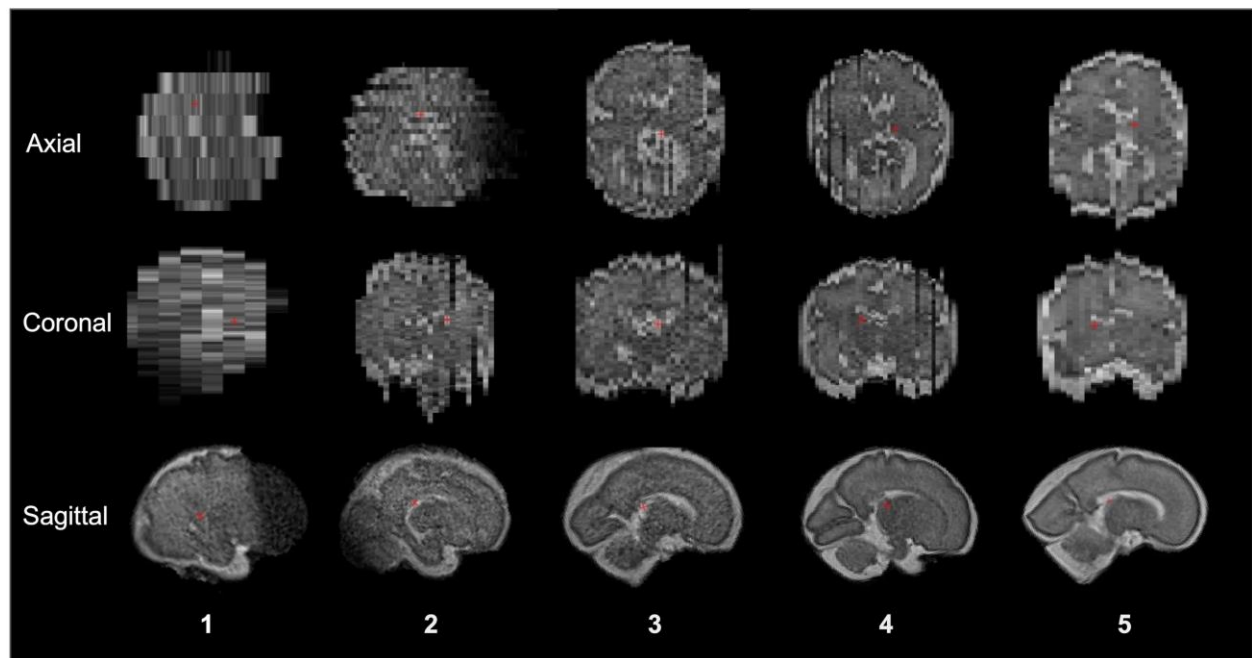


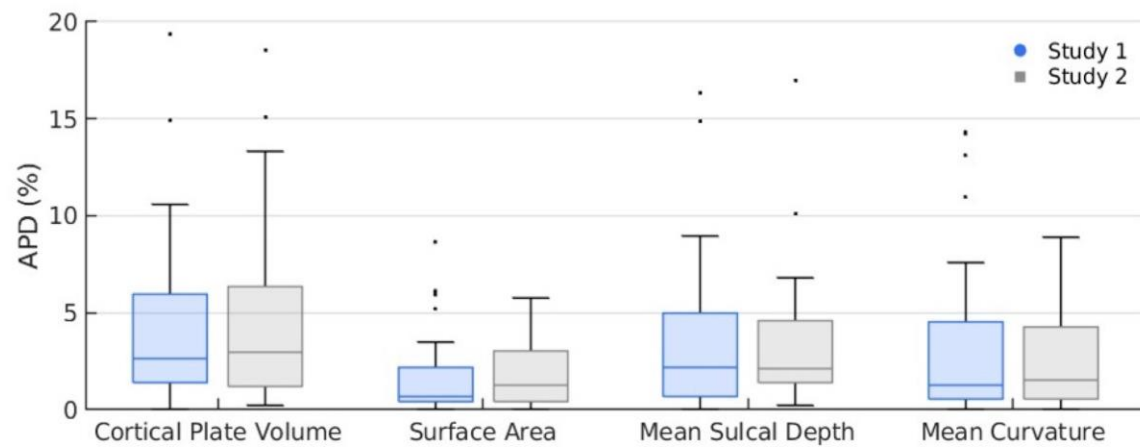
Supplementary Figures



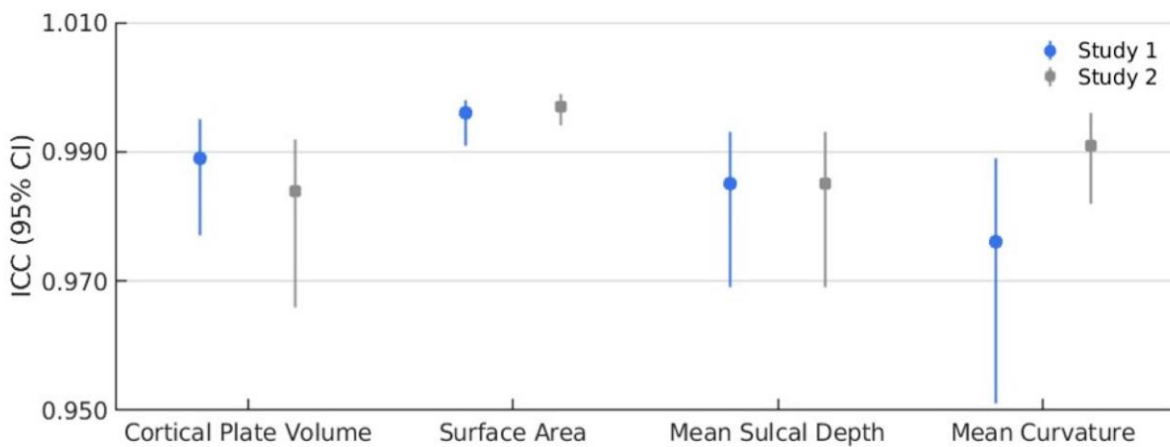
Supplementary Figure 1.

Representative fetal brain MRI slices illustrating the HASTE stack five-point quality scale (from 1: unusable to 5: excellent quality).

[A]



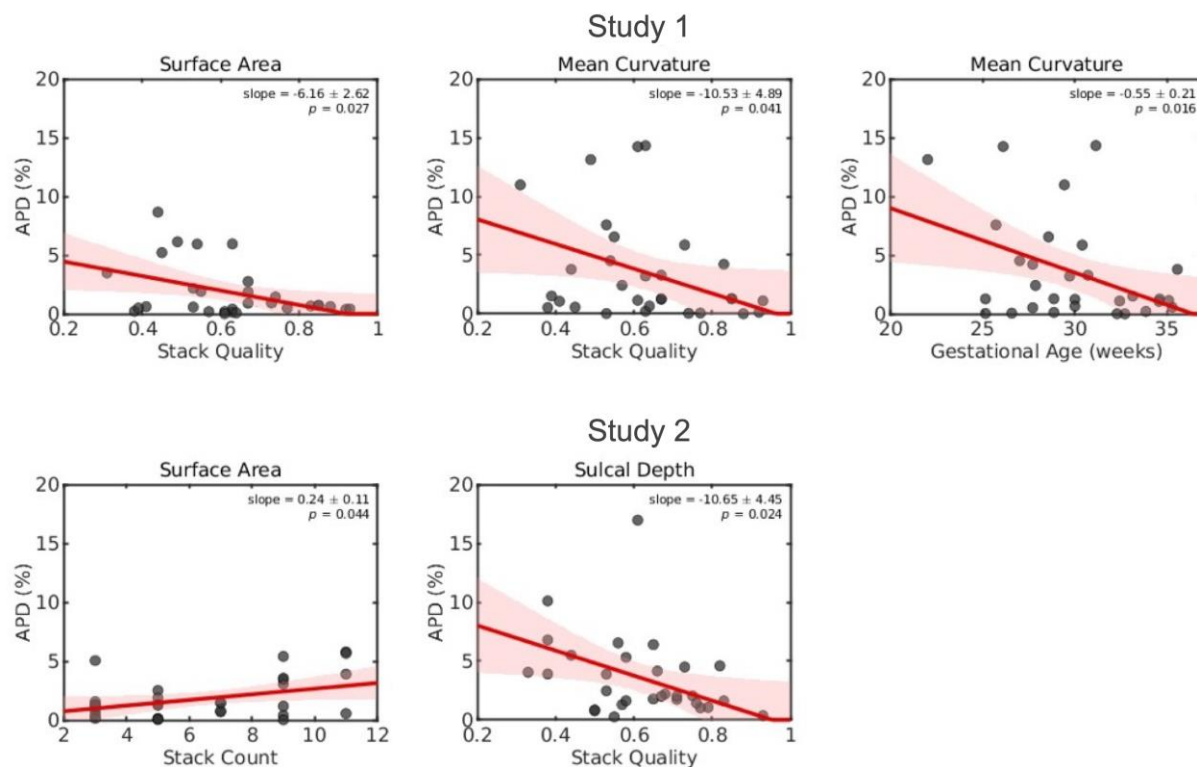
[B]



Supplementary Figure 2. Test-retest reliability of SVRTK-derived cortical metrics.

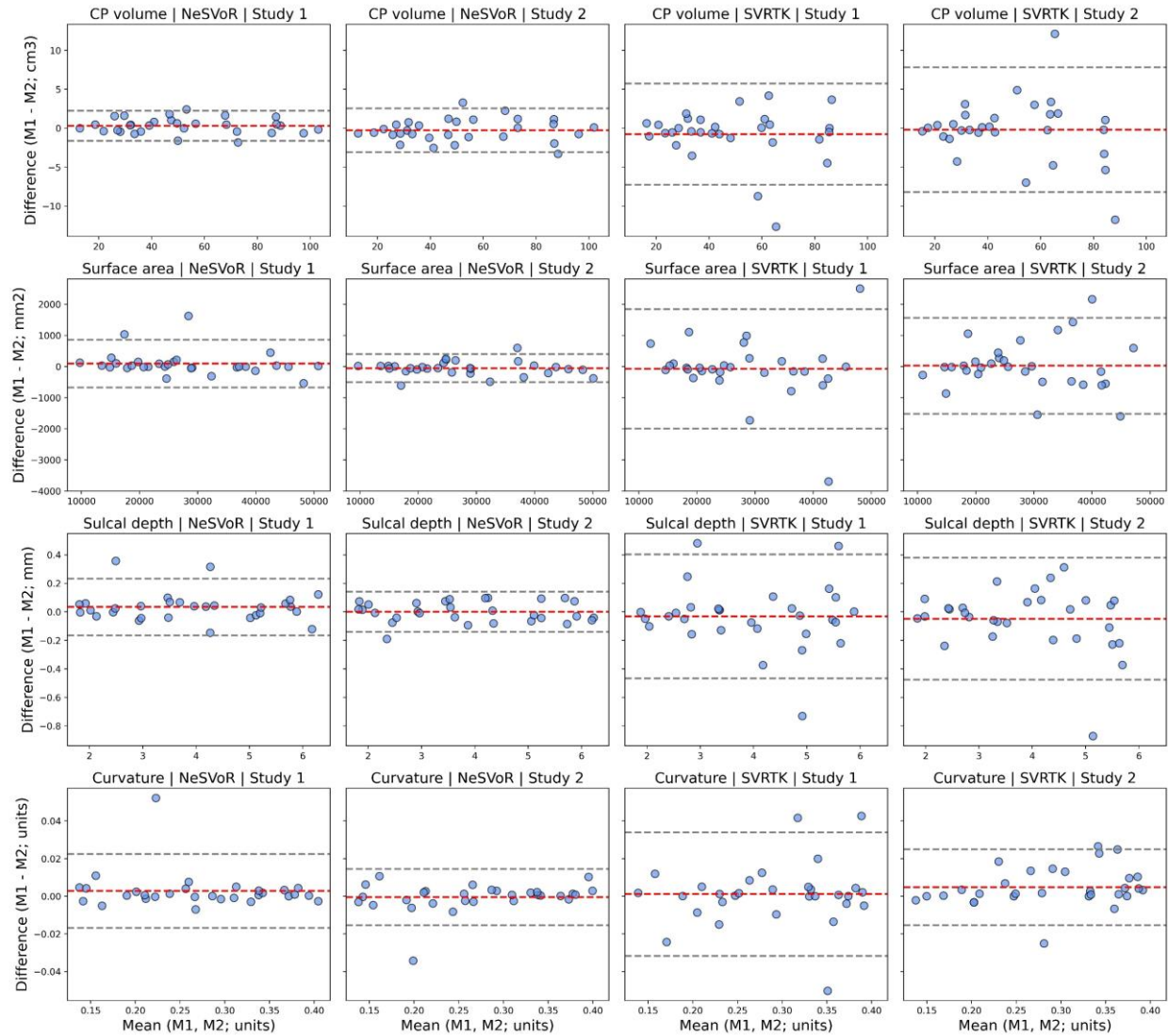
(A) Box plots display APD (%) across subjects for cortical plate volume, surface area, mean sulcal depth, and mean curvature in Study 1 and Study 2.

(B) ICC(2,1) estimates with error bars indicating 95% CIs for cortical plate volume, surface area, mean sulcal depth, and mean curvature in Study 1 and Study 2 (dashed line at ICC = 1).



Supplementary Figure 3. Test-retest reliability of SVRTK-based cortical metrics with imaging factors and GA.

Scatter plots show APD (%) for cortical metrics (y-axis) against mean stack quality, stack count or GA (x-axis) for Study 1 (top) and Study 2 (bottom). Points represent individual subjects, regression lines are displayed for covariate–metric pairs with statistically significant associations (pre-FDR $p < 0.05$), and shaded bands indicate the 95% CIs of the regression fits.



Supplementary Figure 4. Bland–Altman analysis summarizing absolute measure agreement.

Plots show agreement between paired reconstructions across cortical metrics (rows: cortical plate volume, surface area, mean sulcal depth, mean curvature). Columns represent reconstruction method (NeSVoR, SVRTK) and within-session split (Study 1 and Study 2). The solid red line indicates mean bias; dashed lines indicate ± 1.96 standard deviations (limits of agreement). Axes are consistent within measures to enable direct visual comparison.

Supplementary Tables

Supplementary Table 1. Bootstrap stability of multiple linear regression coefficients for APD (NeSVoR and SVRTK reconstructions, Study 1 and Study 2). Values indicate directional consistency (% of bootstrap iterations retaining the same sign as the full-sample estimate) across 1000 subject-level resamples for each predictor and measure.

NeSVoR						SVRTK					
Study	Measure	Mean Stack Quality (%)	Quality Difference (%)	Stack Count (%)	GA (%)	Study	Measure	Mean Stack Quality (%)	Quality Difference (%)	Stack Count (%)	GA (%)
Study 1 (S1–S2)	Cortical Plate Volume	93	52	92	97	Study 1 (S1–S2)	Cortical Plate Volume	90	51	98	70
	Surface Area	85	56	88	91		Surface Area	98	55	67	50
	Mean Sulcal Depth	89	51	99	95		Mean Sulcal Depth	97	51	77	77
	Mean Curvature	96	50	92	100		Mean Curvature	99	96	82	87
Study 2 (S3–S4)	Cortical Plate Volume	64	51	54	93	Study 2 (S3–S4)	Cortical Plate Volume	80	52	90	79
	Surface Area	93	54	96	58		Surface Area	72	51	100	98
	Mean Sulcal Depth	99	52	79	98		Mean Sulcal Depth	79	51	72	58
	Mean Curvature	99	52	83	100		Mean Curvature	89	52	98	74

Supplementary Table 2. Multiple linear regression results of APD (NeSVoR) with mean stack quality, quality difference, mean slice thickness, thickness difference, stack count, and GA.

Regression coefficients ($\beta \pm \text{SE}$, unstandardized) and corresponding p-values (pre-FDR, in parentheses) are shown for each covariate. An asterisk (*) denotes statistical significance at uncorrected $p < 0.05$.

Study	Biometric Measure	Mean Stack Quality ($\beta \pm \text{SE}$, p)	Quality Difference ($\beta \pm \text{SE}$, p)	Mean Slice Thickness ($\beta \pm \text{SE}$, p)	Thickness Difference ($\beta \pm \text{SE}$, p)	Stack Count ($\beta \pm \text{SE}$, p)	GA ($\beta \pm \text{SE}$, p)
Study 1 (S1–S2)	Cortical Plate Volume	-3.60 $\pm$ 1.65 (0.040*)	5.34 $\pm$ 3.25 (0.114)	1.10 $\pm$ 0.45 (0.023*)	0.42 $\pm$ 0.82 (0.617)	-0.23 $\pm$ 0.10 (0.034*)	-0.21 $\pm$ 0.07 (0.008*)
	Surface Area	-1.91 $\pm$ 1.90 (0.324)	-3.93 $\pm$ 3.74 (0.304)	0.64 $\pm$ 0.52 (0.231)	-0.16 $\pm$ 0.95 (0.866)	-0.19 $\pm$ 0.12 (0.115)	-0.11 $\pm$ 0.08 (0.201)
	Mean Sulcal Depth	-5.13 $\pm$ 3.34 (0.139)	-3.29 $\pm$ 6.59 (0.623)	1.19 $\pm$ 0.92 (0.208)	0.12 $\pm$ 1.67 (0.945)	-0.55 $\pm$ 0.20 (0.013*)	-0.26 $\pm$ 0.15 (0.093)
	Mean Curvature	-10.64 $\pm$ 5.11 (0.049*)	-5.11 $\pm$ 10.08 (0.617)	1.70 $\pm$ 1.40 (0.236)	-0.60 $\pm$ 2.56 (0.815)	-0.68 $\pm$ 0.31 (0.039*)	-0.57 $\pm$ 0.23 (0.019*)
Study 2 (S3–S4)	Cortical Plate Volume	-0.68 $\pm$ 2.81 (0.811)	6.70 $\pm$ 5.57 (0.241)	-0.37 $\pm$ 0.81 (0.653)	0.56 $\pm$ 1.05 (0.600)	-0.01 $\pm$ 0.14 (0.945)	-0.21 $\pm$ 0.11 (0.068)
	Surface Area	-1.48 $\pm$ 1.08 (0.183)	0.01 $\pm$ 2.13 (0.996)	0.19 $\pm$ 0.31 (0.540)	0.05 $\pm$ 0.40 (0.895)	-0.12 $\pm$ 0.06 (0.048*)	-0.02 $\pm$ 0.04 (0.707)
	Mean Sulcal Depth	-3.82 $\pm$ 2.26 (0.105)	-2.57 $\pm$ 4.49 (0.573)	-0.16 $\pm$ 0.65 (0.811)	0.32 $\pm$ 0.85 (0.710)	-0.16 $\pm$ 0.65 (0.811)	-0.16 $\pm$ 0.09 (0.095)
	Mean Curvature	-9.49 $\pm$ 4.48 (0.045*)	-5.05 $\pm$ 8.88 (0.575)	0.52 $\pm$ 1.29 (0.693)	-1.10 $\pm$ 1.67 (0.516)	-0.46 $\pm$ 0.23 (0.058)	-0.39 $\pm$ 0.18 (0.038*)

Supplementary Table 3. Test-retest reliability of SVRTK-derived quantitative cortical measurements.

A. Absolute percent difference (APD, mean $\pm$ SD), intraclass correlation coefficients [ICC(2,1), 95% CI], and repeatability coefficients [RC, 95% CI] for cortical plate volume, surface area, mean sulcal depth, and mean curvature in Study 1 (S1–S2) and Study 2 (S3–S4). RC values are reported in cm³ for cortical plate volume, mm² for surface area, mm for mean sulcal depth, and curvature units for mean curvature.

B. Paired statistical comparisons of absolute percent difference (APD) between NeSVoR and SVRTK reconstructions across cortical metrics and studies. Reported are t-statistics from paired t-tests, corresponding false discovery rate (FDR)-corrected p-values, and effect sizes (Cohen's d). Negative t-values indicate lower APD for NeSVoR relative to SVRTK.

A.

Biometric Measure	Study 1			Study 2		
	Mean APD (%) $\pm$ SD	ICC [95% CI]	RC [95% CI]	Mean APD (%) $\pm$ SD	ICC [95% CI]	RC [95% CI]
Cortical Plate Volume	4.07 $\pm$ 4.43	0.989 [0.977, 0.995]	6.49 [4.54, 12.84]	4.84 $\pm$ 4.88	0.984 [0.966, 0.992]	8.01 [6.34, 15.09]
Surface Area	1.81 $\pm$ 2.28	0.996 [0.991, 0.998]	1921.0 [1148.0, 3944.1]	1.84 $\pm$ 1.81	0.997 [0.994, 0.999]	1542.3 [1406.2, 2747.8]
Mean Sulcal Depth	3.60 $\pm$ 4.15	0.985 [0.969, 0.993]	0.44 [0.36, 0.82]	3.58 $\pm$ 3.44	0.985 [0.969, 0.993]	0.43 [0.34, 0.86]
Mean Curvature	3.54 $\pm$ 4.37	0.976 [0.951, 0.989]	0.03 [0.03, 0.06]	2.52 $\pm$ 2.73	0.991 [0.982, 0.996]	0.02 [0.02, 0.04]

B.

		T	p (corr.)	Cohen-D
Cortical Plate Volume	Study 1	-2.55	0.034	0.70
	Study 2	-2.54	0.034	0.63
Surface Area	Study 1	-2.26	0.051	0.52
	Study 2	-3.42	0.015	0.90
Mean Sulcal Depth	Study 1	-1.95	0.081	0.46
	Study 2	-2.81	0.034	0.71
Mean Curvature	Study 1	-1.50	0.165	0.35
	Study 2	-0.79	0.438	0.19

Supplementary Table 4. Multiple linear regression results of APD (SVRTK) with mean stack quality, quality difference, stack count, and GA.

Regression coefficients ($\beta \pm SE$, unstandardized) and corresponding p-values (pre-FDR, in parentheses) are shown for each covariate. An asterisk (*) denotes statistical significance at uncorrected $p < 0.05$.

Study	Biometric Measure	Mean Stack Quality ($\beta \pm SE$, p)	Quality Difference ($\beta \pm SE$, p)	Stack Count ($\beta \pm SE$, p)	GA ($\beta \pm SE$, p)
Study 1 (S1–S2)	Cortical Plate Volume	-5.12 $\pm$ 5.39 (0.351)	-11.69 $\pm$ 11.12 (0.303)	0.24 $\pm$ 0.31 (0.457)	-0.11 $\pm$ 0.23 (0.650)
	Surface Area	-6.16 $\pm$ 2.62 (0.027*)	-6.42 $\pm$ 5.40 (0.245)	-0.22 $\pm$ 0.15 (0.166)	0.01 $\pm$ 0.11 (0.979)
	Mean Sulcal Depth	-8.52 $\pm$ 4.77 (0.086)	-17.94 $\pm$ 9.84 (0.080)	-0.21 $\pm$ 0.28 (0.462)	-0.19 $\pm$ 0.21 (0.376)
	Mean Curvature	-10.53 $\pm$ 4.89 (0.041*)	5.47 $\pm$ 10.09 (0.593)	-0.09 $\pm$ 0.28 (0.750)	-0.55 $\pm$ 0.21 (0.016*)
Study 2 (S3–S4)	Cortical Plate Volume	-6.76 $\pm$ 6.02 (0.272)	-6.77 $\pm$ 12.48 (0.592)	0.61 $\pm$ 0.33 (0.074)	0.21 $\pm$ 0.25 (0.401)
	Surface Area	-4.17 $\pm$ 2.08 (0.056)	5.47 $\pm$ 4.32 (0.217)	0.24 $\pm$ 0.11 (0.044*)	0.06 $\pm$ 0.09 (0.458)
	Mean Sulcal Depth	-10.65 $\pm$ 4.45 (0.024*)	1.61 $\pm$ 9.23 (0.863)	-0.41 $\pm$ 0.24 (0.100)	-0.01 $\pm$ 0.18 (0.978)
	Mean Curvature	-6.81 $\pm$ 3.43 (0.058)	2.02 $\pm$ 7.12 (0.779)	0.22 $\pm$ 0.19 (0.254)	-0.15 $\pm$ 0.14 (0.290)

Supplementary Table 5. Multiple linear regression results of APD (SVRTK) with mean stack quality, quality difference, mean slice thickness, thickness difference, stack count, and GA.

Regression coefficients ($\beta \pm SE$, unstandardized) and corresponding p-values (pre-FDR, in parentheses) are shown for each covariate. An asterisk (*) denotes statistical significance at uncorrected $p < 0.05$.

Study	Biometric Measure	Mean Stack Quality ($\beta \pm SE$, p)	Quality Difference ($\beta \pm SE$, p)	Mean Slice Thickness ($\beta \pm SE$, p)	Thickness Difference ($\beta \pm SE$, p)	Stack Count ($\beta \pm SE$, p)	GA ($\beta \pm SE$, p)
Study 1 (S1–S2)	Cortical Plate Volume	-3.58 $\pm$ 5.33 (0.508)	-11.20 $\pm$ 10.82 (0.311)	-2.79 $\pm$ 1.54 (0.082)	2.05 $\pm$ 2.78 (0.469)	0.30 $\pm$ 0.33 (0.385)	0.15 $\pm$ 0.24 (0.529)
	Surface Area	-5.72 $\pm$ 2.64 (0.041*)	-5.96 $\pm$ 5.35 (0.277)	-1.25 $\pm$ 0.76 (0.114)	0.22 $\pm$ 1.38 (0.873)	-0.16 $\pm$ 0.16 (0.335)	0.04 $\pm$ 0.12 (0.762)
	Mean Sulcal Depth	-8.47 $\pm$ 4.91 (0.098)	-16.81 $\pm$ 9.97 (0.105)	-1.65 $\pm$ 1.42 (0.255)	-1.20 $\pm$ 2.56 (0.644)	-0.07 $\pm$ 0.31 (0.815)	-0.11 $\pm$ 0.22 (0.604)
	Mean Curvature	-10.75 $\pm$ 5.14 (0.048*)	4.86 $\pm$ 10.44 (0.646)	1.15 $\pm$ 1.38 (0.448)	0.32 $\pm$ 2.68 (0.905)	-0.16 $\pm$ 0.32 (0.615)	-0.59 $\pm$ 0.23 (0.017*)
Study 2 (S3–S4)	Cortical Plate Volume	-8.91 $\pm$ 6.32 (0.172)	-4.44 $\pm$ 13.54 (0.746)	0.93 $\pm$ 1.98 (0.642)	-2.88 $\pm$ 2.54 (0.269)	0.60 $\pm$ 0.33 (0.086)	0.24 $\pm$ 0.27 (0.378)
	Surface Area	-4.73 $\pm$ 2.21 (0.043*)	5.65 $\pm$ 4.74 (0.245)	0.08 $\pm$ 0.69 (0.911)	-0.91 $\pm$ 0.89 (0.314)	0.24 $\pm$ 0.11 (0.044*)	0.08 $\pm$ 0.09 (0.385)
	Mean Sulcal Depth	-10.41 $\pm$ 4.79 (0.040*)	-0.57 $\pm$ 10.28 (0.957)	-0.85 $\pm$ 1.50 (0.576)	-0.48 $\pm$ 1.93 (0.805)	-0.39 $\pm$ 0.25 (0.140)	-0.04 $\pm$ 0.20 (0.840)
	Mean Curvature	-6.60 $\pm$ 3.72 (0.089)	1.40 $\pm$ 7.98 (0.862)	-0.24 $\pm$ 1.17 (0.837)	0.11 $\pm$ 1.50 (0.945)	0.22 $\pm$ 0.20 (0.268)	-0.14 $\pm$ 0.16 (0.374)